Assignment - 3

Answer the following Questions :-

1. Difference b/w AI & Generative AI.

Artificial Intelligence (A.I) is a branch of computer science dedicated to creating machines capable of intelligent behaviour. This includes problem-solving, recognizing speech, translating languages and decision-making.

Generative A.I (~~Gen~~ Gen AI) represents next step in evolution of AI. It's a subset of AI focused on creating new and original content. Unlike traditional AI systems which operate based on pre-existing data & ideas, from digital art to novel text compositions. It uses advanced machine learning techniques like deep learning, to not just analyze but also produce content that is creative

and contextually relevant.

Artificial Intelligence	Generative AI
→ Analyzing data, automating processes and making decisions based on existing information.	→ Creating new, original content by generating data & ideas.
→ Decision trees, pattern recognition, predictive modeling.	→ Deep learning, neural networks, creative data generation.
→ Relies on structured data for specific tasks.	→ Utilizes both structured and unstructured data
→ Predictive analytics, fraud detection, personalized recommendations, process automation.	→ Automated content creation, AI generative art, Synthetic data generation, content moderation.

3. Discuss all the AI Regulations & Laws.

Global governance of artificial intelligen relies on a combination of legally binding frameworks, federal executive policies and voluntary international safety standards designed to protect human rights while fostering technological innovation.

The World's first comprehensive legal frameworks is the European Union Artificial Intelligence Act (EU AI act). It uses a risk-based classification system:

→ Unacceptable Risk: Strictly bans harmful applications (e.g. Social Scoring or cognitive manipulation)

→ High Risk: Imposes stringent safety data governance and human oversight obligations on critical tools (eg. hiring algorithms

medical devices, public infrastructure)

→ Limited Risk: Requires transparency, such as disclosures for chatbots and AI-generated deepfakes

→ Minimal Risk: Remains unregulated.

3. What is the role of A.I in Health care?

Artificial Intelligence plays a transformative role in health care by streamlining & administrative workflows, enhancing diagnostic accuracy and personalizing patient care. AI systems process vast volumes of clinical data, such as electronic medical records, genomic data, vital signs, and laboratory results to assist healthcare professionals across all clinical stages.

Key applications include:

- Early Disease Detection: Identifying subtle risk indicators before symptoms appear.
- Workflow Automation: Reducing administrative burdens by generating clinical notes, handling scheduling and triaging patient communications.
- Drug Discovery: Accelerating identification and testing of novel therapeutic compounds.
- Personalized Medicine: Tailoring treatment regimens based on an individual patient's unique biological and genetic profile.

By acting as an intelligent assistant AI enables clinicians to make faster data-driven decisions, reduce human error and allocate more time to direct patient care and communications.

4. Discuss benefits and challenges of AI in Healthcare.

Integration of AI into medical practice provides substantial benefits:

- Increased Diagnostic Precision: Enables earlier and more accurate disease identification.
- Speed & Efficiency: Accelerates image analysis and patient triaging, saving vital times in emergency settings.
- Cost Reduction & Scalability: Automates repititive administrative tasks, lowering healthcare overhead and reducing diagnostic error rates.

However clinical adoption faces major challenges:

- Data Privacy & Security: Safeguarding sensitive patient health information against breaches

- Algorithmic Bias : Potential disparities if AI training datasets lacks demographic diversity
- The "Black box" Problem: A lack of transparency in how complex algorithms reach specific clinical conclusions.
- Liability & Regulations.: Unclear legal accountability for diagnostic errors and high costs associated with system integration and regulatory compliances

5. Explain AI Imaging

AI imaging refers to using computing vision and deep learning algorithms to process, evaluate, interpret radiological medical scans — including X-rays computed Tomography scans, Magnetic Resonance Imaging (MRI) & ultrasound

These specialized neural networks trained on millions of annotated radiological files to identify complex visual patterns. They can detect microscopic tissue lesions, spot early stage tumors, identify bone microfracture, track organ or tumor dimensions over time by exceptional precision.

In clinical radiology, AI imaging serves two primary functions:

1. Quality Improvement: Reconstructing high-resolution images rapidly, reducing overall scan times and patient anxiety.

2. Triage support: Automatically prioritizing high-risk scans (eg. suspected acute stroke or internal hemorrhaging) for immediate radiologist evaluation.